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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 14 Controlling the position of a cart

Sensors and Instrumentation

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1. Objective

Develop a position control system for a motorized cart, using a
an ultrasonic sensor and a PID controller implemented on Arduino, in order to
maintain a fixed distance from a wall.

2. Introduction

In the field of mechatronics and automatic control, one of the fundamental challenges
It is about ensuring that a system maintains a desired position or trajectory in the face of potential
External disturbances. In this exercise, a position control system is implemented.
for a cart with direct current motors, whose objective is to maintain a distance
constant with respect to a wall using an HC-SR04 ultrasonic sensor.
The system is controlled by a PID (Proportional, Integral, Derivative) algorithm.
which dynamically adjusts the speed and direction of the cart to minimize error
between the current and desired distance. The latter is established remotely through
via a Bluetooth interface, allowing real-time experimentation with different values
reference.
The use of PID control in positioning systems is widely used in industry
Due to its simplicity and effectiveness, this practice allows the student to understand
to practically explain the operation of a closed-loop controller, as well as the
importance of properly adjusting its parameters to obtain a stable response and
precise.

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3. Definitions and conceptual framework

3.1. Ultrasonic sensor

This sensor allows measuring distances through the time of flight of an ultrasonic pulse.

By sending a high-frequency sound pulse and detecting the echo reflected off an object, it is

It is possible to calculate the distance using the formula:

$$\text{distance} = 2 \quad \frac{340\ \text{m/s} \cdot t(\mu s)}{} \cdot \frac{1\ \text{s}}{1{,}000{,}000\ \mu s} \cdot \frac{100\ \text{cm}}{1\ \text{m}}$$

Which, simplified, becomes

$$\text{distance (cm)} = t(\mu s) \times 0.01715$$



Figure 1: HC-SR04 Ultrasonic Sensor

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3.2. PID Control

The PID (Proportional, Integral, Derivative) controller is one of the most used in system control. Its output is based on three components:

- Proportional (P): responds proportionally to the current error.
- Integral (I): accumulates the error over time to correct persistent errors.
- Derivative (D): anticipates the error trend to reduce oscillations.

The combination of these three actions allows for a balance between speed and stability. and accuracy. In this case, the PID is implemented with the ARDUINO PID v1 library in the ESP 32, configuring the parameters empirically until a behavior is achieved desired.

3.3. Bluetooth Communication

Bluetooth communication allows data to be sent and received wirelessly. In this In practice, the ESP32 integrated module is used along with the BluetoothSerial library. to receive the setpoint value in real time from an external device, allowing modify the position reference without reprogramming the system using a Android application created in MIT App Inventor.

3.4. Direct Current (DC) Motors

Direct current motors convert electrical energy into rotary motion. They are widely used due to their ease of control and low cost. In this practice, They used 6V motors, which allow for acceptable speeds for a laboratory or classroom environment.

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Figure 2: 6V DC Motor

3.5. PWM Signals

Pulse width modulation (PWM) is a technique that allows the speed of a DC motor to be controlled by adjusting the duty cycle.
cycle) of a digital signal. The higher the duty cycle, the greater the energy supplied to the engine, and therefore, its speed.

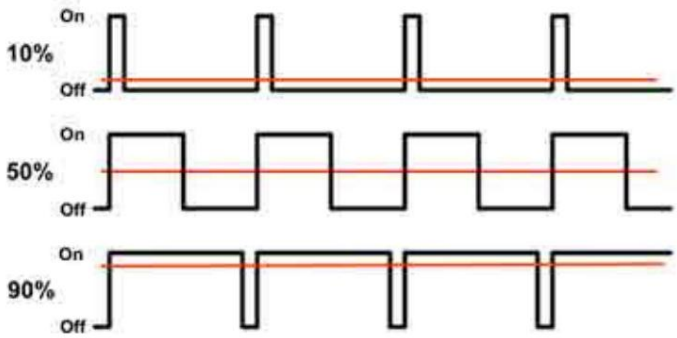


Figure 3: PWM Graph

3.6. Bridge H (L298N)

The H-bridge is a circuit that allows changing the polarity of the power supply to the motors, which allows their direction of rotation to be reversed. The L298N is a dual-bridge motor that

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It allows you to control two motors independently, and can also control their speed using the PWM signal sent from a microcontroller such as the Arduino.

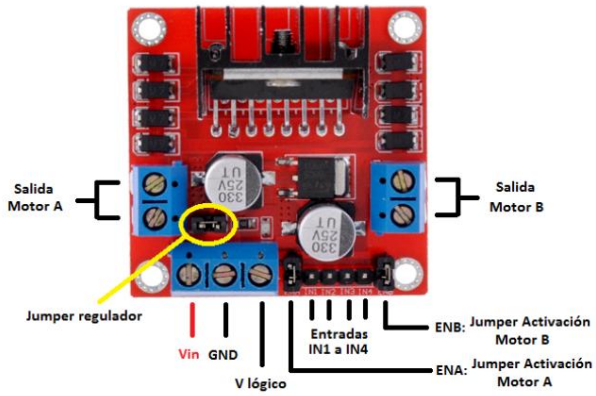


Figure 4: H-bridge L298N

4. Development

The position control system was implemented on an ESP32 board, which was responsible for both from the reading of the HC-SR04 ultrasonic sensor and from the control of the motors through the L298N H-bridge. The programming was done in the Arduino environment for Controlling an ESP32, using the PID_v1.h library to implement the controller PID and the BluetoothSerial.h library to receive data from a mobile device or computer via Bluetooth.

The program flow is as follows:

- Initialization of serial and Bluetooth communication.
- Pin configuration for motor control and ultrasonic sensor.
- Waits for a setpoint value sent via Bluetooth, which represents the distance desired distance in centimeters between the cart and the wall.
- Constant distance reading using the time of flight of the ultrasonic sensor.
- Calculation of the control signal using the PID algorithm.

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- Application of the control signal to the motors, using PWM to adjust the speed according to the magnitude of the error.
- If the calculated PWM is very low (but not zero), a minimum value is imposed for to ensure that the engines respond.

The code was adjusted empirically, mainly the values of the constants of the PID and the minimum PWM threshold needed for the cart to move, because with values At very low speeds, the cart did not move. It was observed that, with the constants Kp = 1.0 and Ki = 0.0 and Kd = 0.2, the system's behavior was sufficiently stable and responded appropriate form for position corrections.

5. Code

This was the code used, as explained above:

Listing 1: PID control code for the cart

```
# include < PID_v1.h >
#include          "BluetoothSerial .h"
BluetoothSerial SerialBT;

// L298N motor pins
const int M1_A = 16                ,    M1_B = 17;
const int M2_A = 18                ,    M2_B = 19;

// HC - SR04 Pins
const int trigPin = 27                    ,    echoPin = 26;
long duration;
double readDistance() {
    digitalWrite(trigPin,                 LOW ) ;
    delayMicroseconds (2) ; // Optional, but can help stabilize
    digitalWrite(trigPin,                 HIGH);
    delayMicroseconds (10) ;
    digitalWrite(trigPin,                 LOW ) ;
    long duration = pulseIn ( echoPin                    ,    HIGH); // no timeout
    return duration * 0.01715;
```

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```
}

// PID
double Setpoint , Input , Output ;
double Kp = 1.0 , Ki = 0.0 , Kd = 0.2;
PID myPID (& Input , & Output, & Setpoint, Kp, Ki , Kd , DIRECT);

// Minimum motor threshold
const int motorMinPWM = 135;

void setup() {
    Serial . begin (115200) ;
    while (! Serial ) delay (10) ;
    Serial.println ("\n--- Distance PID Control ---") ;

    SerialBT . begin (" ESP32_Carrito ") ; // Bluetooth Name

    // Motor pins
    pinMode ( M1_A , OUTPUT ) ; pinMode ( M1_B , OUTPUT ) ;
    pinMode ( M2_A , OUTPUT ) ; pinMode ( M2_B , OUTPUT ) ;
    // Ultrasonic pins
    pinMode (trigPin , OUTPUT ); pinMode ( echoPin , INPUT ) ;

    // Wait for first Bluetooth Setpoint
    Serial. println ("Waiting for Bluetooth Setpoint (1 byte in cm)... ");
    while ( SerialBT . available () == 0) delay (10) ;
    Setpoint = SerialBT . read () ;
    Serial. print ("Initial setpoint received: ") ;
    Serial . print ( Setpoint ) ;
    Serial . println (" cm") ;

    // Init PID
    Input = readDistance() ;
    myPID . SetMode(AUTOMATIC);
    myPID . SetOutputLimits ( -255 , 255) ;

    Serial. println ("Setup complete. Starting control... ");
```

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```
}

void loop () {
    // Update Setpoint if a new one arrives
    if ( SerialBT . available () > 0) {
        Setpoint = SerialBT . read () ;
        Serial.print ("New Setpoint received: ");
        Serial . print ( Setpoint ) ;
        Serial . println (" cm") ;
    }

    // 1) Read distance
    Input = readDistance() ;
    Serial. print ( "Current Dist: ") ; Serial. print (Input); Serial. println ("cm" )

    // 2) Calculate PID
    myPID . Compute() ;

    // 3) Motor control
    int pwm = abs (( int ) Output ) ;
    bool forward = (Output > 0) ;

    if ( pwm > 0 && pwm < motorMinPWM ) pwm = motorMinPWM ; // Apply minimum PWM

    if (ahead) {
        analogWrite ( M1_A , pwm ) ; analogWrite ( M2_A , pwm ) ;
        analogWrite ( M1_B , 0) ; analogWrite ( M2_B , 0) ;
    } else {
        analogWrite ( M1_B , pwm ) ; analogWrite ( M2_B , pwm ) ;
        analogWrite ( M1_A , 0) ; analogWrite ( M2_A , 0) ;
    }

    // 4) Debug in serial monitor
    Serial. print ("SP:" ) Serial. print ( Setpoint ) ;
    Serial. print ("PV:") ; Serial. print (Input);
    Serial. print ("Err:") ; Serial. print ( Setpoint - Input ) ;
    Serial. print ("Out:" ) Serial. print ( Output ) ;
```

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```
    Serial. print ("PWM:" ) Serial. println ( pwm ) ;

    delay (100) ;
}
```

6. Results

During system testing, a range of desired distances between 10 cm and 50 cm, dynamically transmitted via Bluetooth. Within this range, the cart maintained a stable position facing the wall, continuously correcting its distance by means of the PID controller.

It was observed that, when the desired distance was increased beyond 50 or 60 cm, the sensor The ultrasonic scale was beginning to show measurement errors. This is probably due to The sensor operates with a detection angle of approximately 30°, so At large distances, it may begin to detect the floor instead of the wall, affecting the reading accuracy.

Within the optimal range (10–50 cm), the system's behavior was smooth, without oscillations. very marked or very abrupt movements. It was also verified that the use of a threshold A minimum PWM (motorMinPWM) was required to prevent the motors from stalling. static with very small signals, especially when the error was low but still It needed correction.

The PID output signal adjusted well to the setpoint changes, allowing the cart adapts to new positions with a quick and stable transition. All the The system's behavior could be monitored in real time from the serial monitor.

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7. Conclusion

The implemented system successfully controlled the position of a cart relative to
to a wall using an ultrasonic sensor and a PID controller. The integration of the
The Bluetooth module allowed dynamic modification of the setpoint, which facilitated experimentation.
with different target distances without needing to modify the code or restart the system.
The ultrasonic sensor was found to be reliable within an approximate range of 10 to 50 cm.
but it begins to fail at greater distances due to its conical detection shape,
which can cause erroneous readings by picking up echoes from the floor or other objects. This highlights
the importance of knowing the physical limitations of the sensors when designing
a control system.

The use of the PID controller allowed for smoother behavior of the cart, adjusting its
position continuously without needing to stop completely. Empirical adjustments to
The constants Kp, Ki, and Kd were key to achieving stable and responsive performance.
Finally, this exercise was useful for reinforcing concepts of automatic control and electronics.
and applied programming, showing how to integrate multiple components into a system
functional.